

Azure Data Factory — Parameterized Pipeline

Pipeline Parameters, Variables & Dynamic Expressions

Project Status: Complete

This project demonstrates parameter-driven variables and a dynamic expression to calculate area in an Azure Data Factory pipeline.

Step 1 — Initial Pipeline

The pipeline starts with Set Variable activities for length, breadth, and Area.

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Data Factory

Validate all

Publish all

Migrate to Fabric (Preview)

Factory Resources

Filter resources by name

Pipelines

pipeline1

Change Data Capture (preview)

Datasets

Data flows

Power Query

pipeline1

Activities

set

Validate the current resource

General

Set variable

length

breadth

Area

Parameters

Variables

Settings

Output

Pipeline 85dfea8b-6c4d-4f32-98f2-ef674fceb3d

run ID 8f2-ef674fceb3d

Pipeline status In progress

All status

Monitor in Azure Metrics

Export to CSV

Showing 1 - 2 of 2 items

Activity name	Activity st...	Activit...	Run start	Du
length	In progress	Set variable	8/18/2026, 12:47:53 PM	2s
breadth	Succeeded	Set variable	8/18/2026, 12:47:53 PM	Les

Cancel

Properties

General

Related

Name *

pipeline1

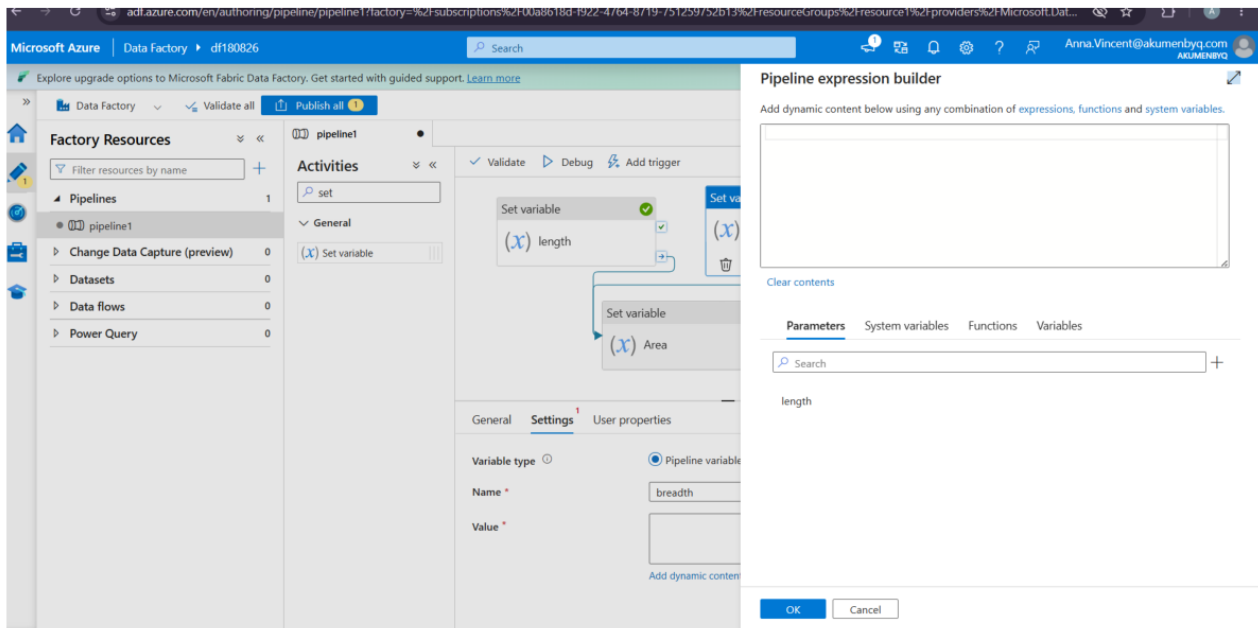
Description

Annotations

New

Step 2 — Expression Builder

The Expression Builder is opened to create a dynamic value for a variable.



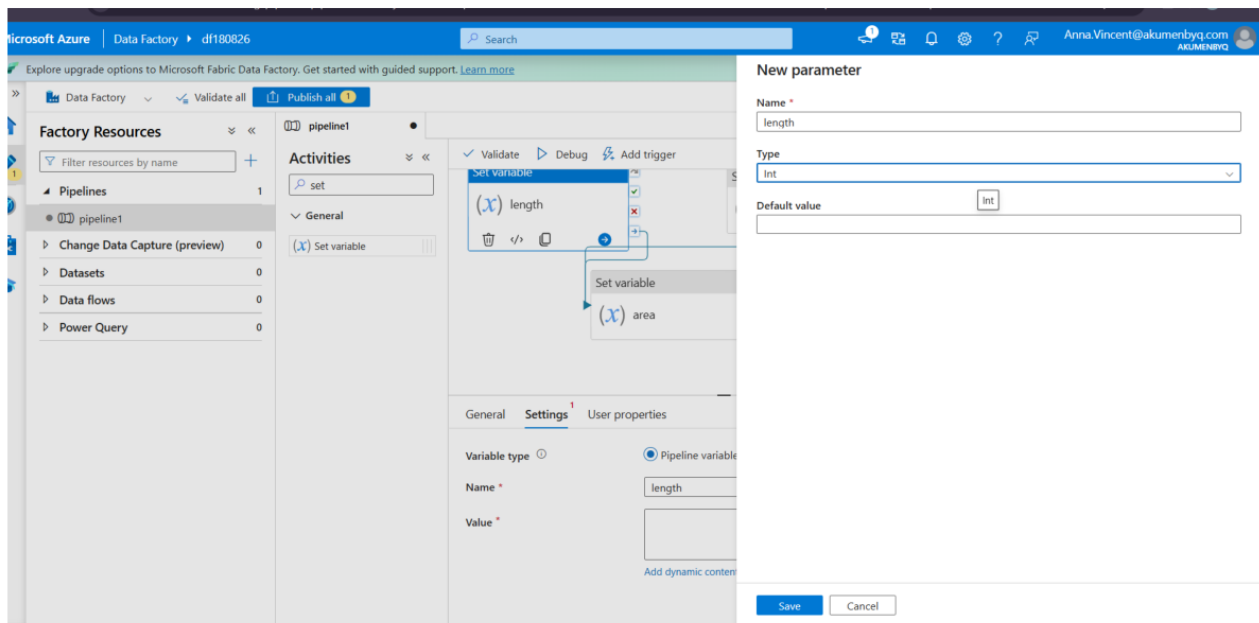
Step 3 — Area Expression

The Area variable is configured using a multiplication expression.

The screenshot displays the Microsoft Azure Data Factory portal. On the left, the 'Factory Resources' pane shows a list of resources including 'pipeline1'. The main workspace shows a pipeline diagram with three 'Set variable' activities. The first two are for 'length' and 'breadth'. The third activity, 'Area', is highlighted with a blue selection box. Below the diagram, the 'Settings' tab for the 'Area' variable is open. It shows 'Variable type' set to 'Pipeline variable', 'Name' set to 'area', and 'Value' set to the expression '@mul(pipeline().parameters.breadth,...)'. The right-hand 'Properties' pane shows the 'General' tab with fields for 'Name' (pipeline1), 'Description', and 'Annotations'.

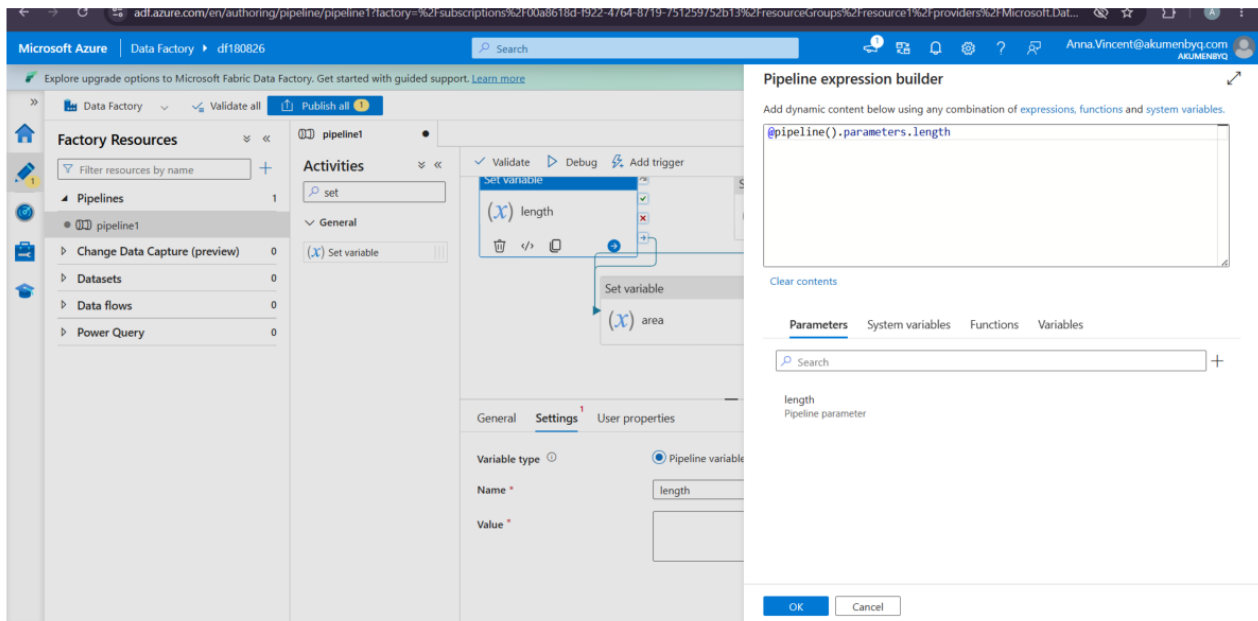
Step 4 — Length Parameter

A pipeline parameter named length is created with the Int data type.



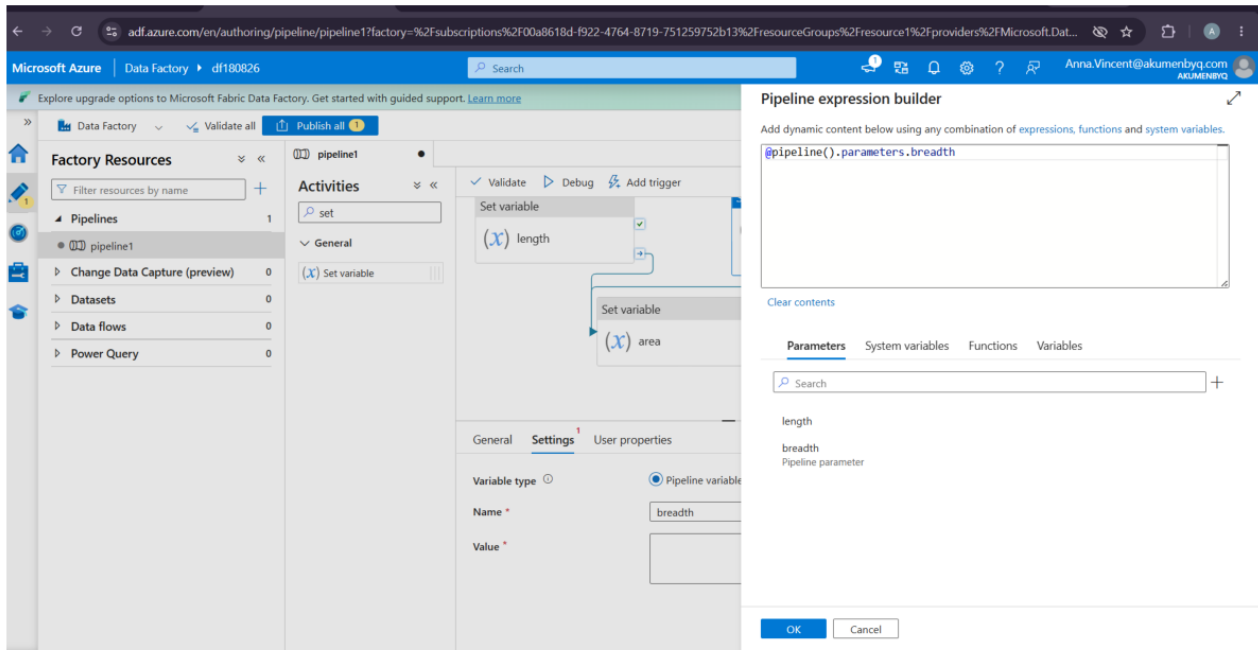
Step 5 — Length Reference

The length variable gets its value from the length pipeline parameter.



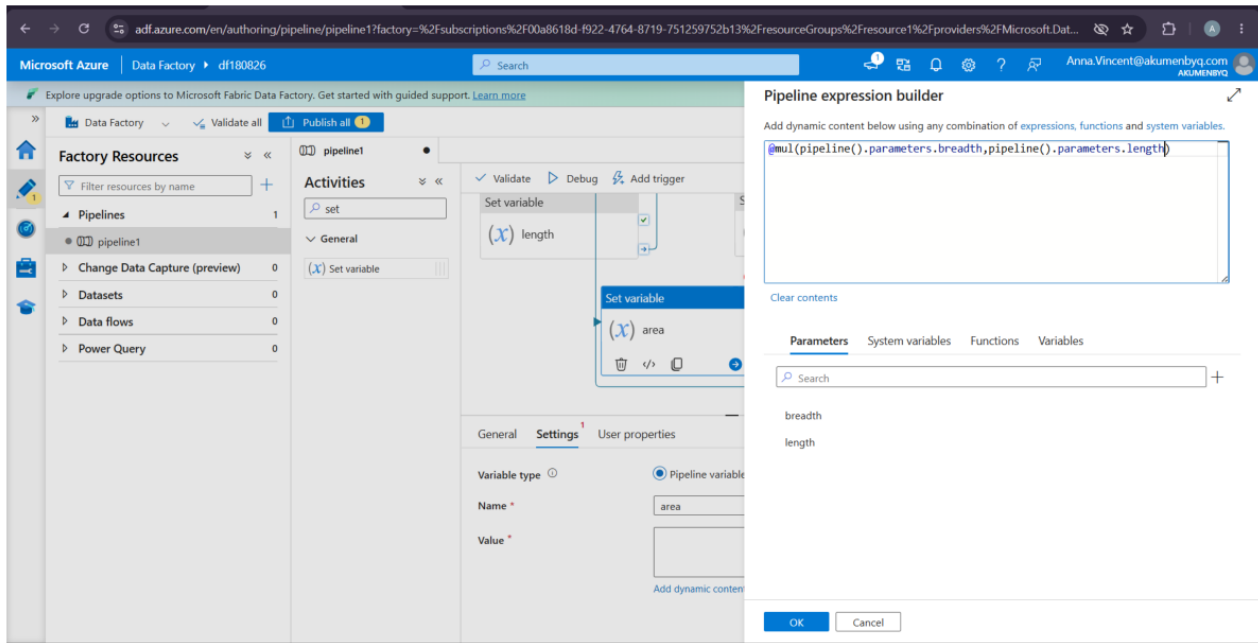
Step 6 — Breadth Reference

The breadth variable gets its value from the breadth pipeline parameter.



Step 7 — Area Calculation

The Area expression multiplies the breadth and length parameters.



Step 8 — Parameterized Pipeline

The completed pipeline shows the parameter-driven Area calculation.

The screenshot displays the Microsoft Azure Data Factory portal interface. The browser address bar shows the URL: `adf.azure.com/en/authoring/pipeline/pipeline1?factory=%2Fsubscriptions%2F00a8618d-f922-4764-8719-751259752b13%2FresourceGroups%2Fresource%2Fproviders%2FMicrosoft.Dat...`. The page title is "Data Factory > df180826".

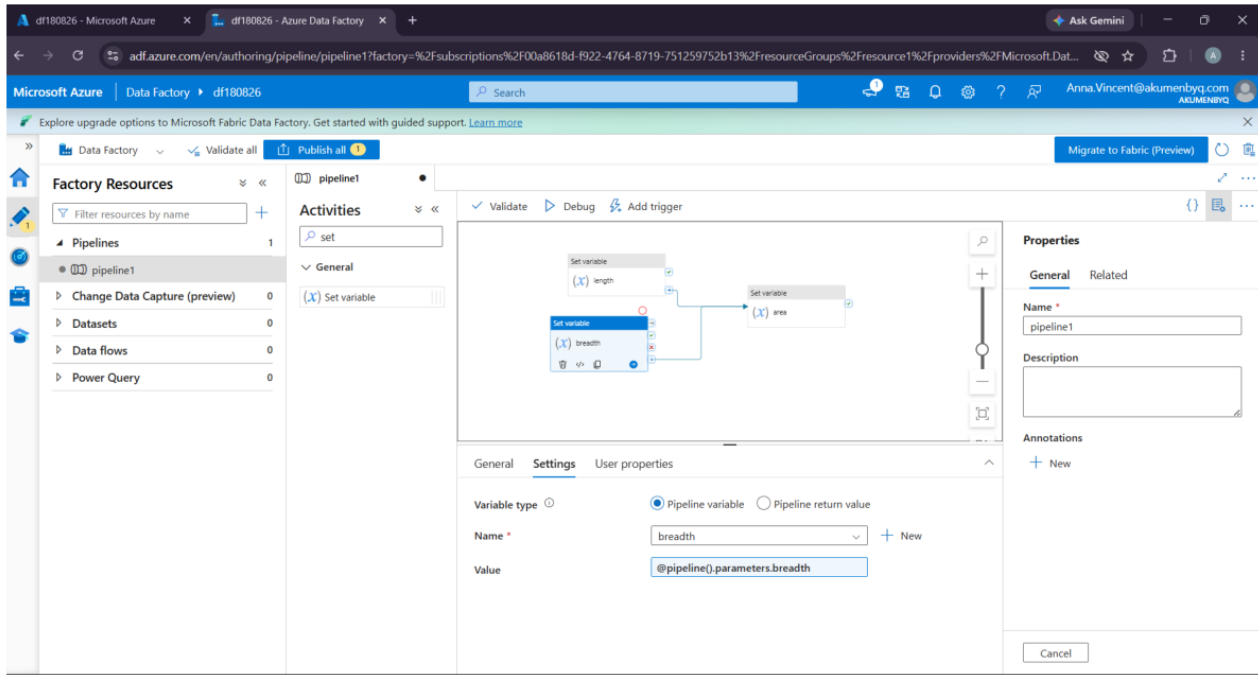
The interface is divided into several sections:

- Factory Resources:** A sidebar on the left showing a tree view with "Pipelines" (1 item), "Change Data Capture (preview)" (0 items), "Datasets" (0 items), "Data flows" (0 items), and "Power Query" (0 items). The "Pipelines" section is expanded, showing "pipeline1".
- Activities:** A central panel showing the pipeline design. It includes a "Set variable" activity with a dropdown menu open, showing options for "(X) length", "(X) breadth", and "(X) area". The "(X) area" option is selected.
- Properties:** A panel on the right showing the properties of the selected activity. The "Name" field is set to "pipeline1". The "Description" field is empty. The "Annotations" section has a "+ New" button.
- Settings:** A panel at the bottom showing the settings for the selected activity. The "Variable type" is set to "Pipeline variable". The "Name" field is set to "area". The "Value" field is set to "@mul(pipeline().parameters.breadth...".

The bottom of the screen shows a Windows taskbar with various application icons and a system tray displaying the temperature (29°C), weather (Partly sunny), and time (13:06, 18-08-2026).

Step 9 — Debug Run

The pipeline is tested with length = 15 and breadth = 10.



Step 10 — Validation

The pipeline is checked and no errors are found.

The screenshot displays the Microsoft Azure Data Factory portal interface. The main area shows the 'pipeline1' in a 'Validating...' state. The 'Activities' pane lists a 'set' activity. The 'Settings' tab for the 'set' activity is active, showing 'Variable type' as 'Pipeline variable', 'Name' as 'breadth', and 'Value' as '@pipeline().param'. A 'Pipeline run' dialog is open on the right, showing parameters for 'length' (Type: Int, Value: 15) and 'breadth' (Type: Int, Value: 10). The dialog has 'OK' and 'Cancel' buttons at the bottom.

Microsoft Azure | Data Factory > df180826

Factory Resources

- Pipelines: 1
 - pipeline1
- Datasets: 0
- Data flows: 0
- Power Query: 0

Activities

- set

General

- (X) Set variable

Settings

Variable type: Pipeline variable

Name: breadth

Value: @pipeline().param

Pipeline run

Parameters

Name	Type	Value
length	Int	15
breadth	Int	10

OK Cancel

29°C Partly sunny

13:07 18-08-2026

Step 11 — Final Result

The Area output is 150, calculated from 15×10 .

The screenshot displays the Microsoft Azure Data Factory portal interface. The top navigation bar shows the URL `adf.azure.com/en/authoring/pipeline/pipeline1?factory=%2Fsubscriptions%2F00a8618d-1922-4764-8719-751259752b13%2FresourceGroups%2Fresource1%2Fproviders%2FMicrosoft.Data...` and the user `Anna.Vincent@akumenbycg.com`. The main interface is divided into several sections:

- Factory Resources:** A sidebar on the left showing a tree view with `Pipelines` (1 item), `Change Data Capture (preview)` (0 items), `Datasets` (0 items), `Data flows` (0 items), and `Power Query` (0 items). The `Pipelines` section is expanded, showing `pipeline1`.
- Activities:** A central panel showing a workflow diagram with two `Set variable` activities. The first activity is named `length` and the second is named `breath`. The `Properties panel on the right shows the General tab for the selected activity, with Name set to pipeline1 and Description empty.`
- Output:** A panel at the bottom showing the pipeline's status and details. It includes the `Pipeline ID` (0989e038-7037-4cd1-5ef-b9b8fef78331), `Pipeline status` (In progress), and a `Showing 1 - 2 of 2 items` section.
- Pipeline validation output:** A panel on the right showing a success message: "Your pipeline has been validated. No errors were found." with a green checkmark icon.

Step 12 — Successful Execution

The final execution shows the pipeline activities completed successfully.

The screenshot displays the Microsoft Azure Data Factory console for a pipeline named 'pipeline1'. The interface is divided into several sections:

- Factory Resources:** A sidebar on the left showing the hierarchy: Pipelines (1), Change Data Capture (preview) (0), Datasets (0), Data flows (0), and Power Query (0).
- Activities:** A central workspace showing the pipeline design. It includes a 'set' trigger, followed by three 'Set variable' activities for 'length', 'breadth', and 'Area'. Each activity has a green checkmark indicating success.
- Output:** A table at the bottom showing the execution results for the three activities.
- Properties:** A panel on the right showing the pipeline's general properties, including the name 'pipeline1' and a description field.

The 'Output' table shows the following data:

Activity name	Activity st...	Activit...	Run start	Du
Area	✓ Succeeded	Set variable	8/18/2026, 12:47:55 PM	Les
length	✓ Succeeded	Set variable	8/18/2026, 12:47:53 PM	2s
breadth	✓ Succeeded	Set variable	8/18/2026, 12:47:53 PM	Les